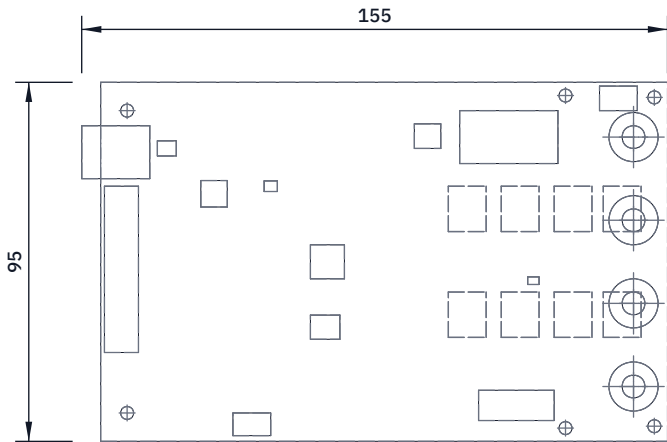


REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-09-30	AC
P2	Status light and light pipe added (CGD-DEC-001)	2026-10-02	AC



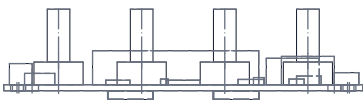
TOP VIEW

Scale 1:2; looking down (along -Z)



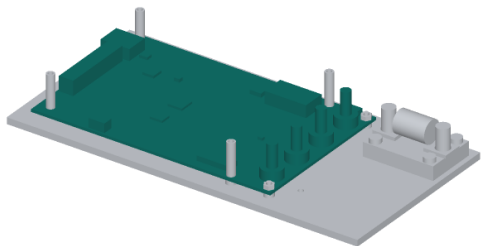
FRONT VIEW

Scale 1:2; looking at the front (along +Y)



RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



WHERE IT GOES

This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Ordered from a board maker; the copper layout is drawn at TRL 4 and must keep every position on this sheet.

Outline 150 x 95 mm, 1.6 mm. Six 3.4 mm unplated holes with a 3 mm copper-free ring, at the plate's six board fixings.

Underside: eight MOSFETs in two rows of four, 14 mm each side of the centre line, 22, 36, 50 and 63 mm right of the board centre.

Top, power end: four M6 stud terminals 66 mm right of centre, B-, P-, P+, B+ at 33 and 11 mm each side (B+ on the far side).

Top, signal end: balance header and CAN and UART connector at the left edge; the CAN connector overhangs the edge by 5 mm.

Top: three-colour status light 30 mm left of the board centre, 20 mm to the B+ side, under the cover's light pipe.

Solder fine-pitch parts on a hot plate, then the MOSFETs from below, then the stud terminals and connectors by hand.

Check: MOSFET tops level within 0.1 mm; stud pins trimmed to 1 mm under the board.

CELLGUARD

github.com/BoujeeEnjinia1701/cellguard

CellGuard main board: outline and fixed positions

DRAWING NO. CGD-DWG-103			REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:2	UNITS mm	PROJECTION 	DATE 2026-10-02		DRAWN Amish Chadha
MATERIAL / NOTES FR-4 1.6 mm, 4 layers, 2 oz copper					
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